

SAMIUL HOSSAIN SAJAL

Cologne, Germany
www.samiulhossain.com
samiulhossainsajal@gmail.com
www.linkedin.com/in/samiul-sajal
https://github.com/SAMIULSAJAL

SUMMARY

Condensed matter physics graduate student with experience in TEM image segmentation, grain tracking, and molecular dynamics simulation. Currently developing computer-vision workflows for materials analysis at Forschungszentrum Jülich (IAS-9) using PyTorch, SAM-based models, and HPC.

RESEARCH & PROFESSIONAL EXPERIENCE

Student Research Assistant (Nov 2025 - Present)

IAS-9, Forschungszentrum Jülich

- Master's thesis focused on segmentation and tracking of in-situ TEM data.
- U-Net, SAM2, SAM3 and MatSAM workflows for TEM image and video analysis
- Grain tracking using overlap analysis and Hungarian matching
- Post-processing, topology refinement and Validation against manually annotated data

Master's Thesis Research (Oct 2019 - Nov 2021)

Shahjalal University of Science & Technology (SUST)

- Molecular Dynamics simulations using ReaxFF and LAMMPS
- HPC-based analysis of aluminum nanoparticle formation and oxidation
- Temperature-dependent growth analysis with scientific visualization

EDUCATION

(Oct 2022 - Present)

M.Sc. in Physics

University of Cologne, Germany.

Primary: Condensed Matter Physics
Secondary: Computational Physics

(Jun 2019 - Nov 2021)

M.Sc. in Physics
SUST, Bangladesh.

(Mar 2015 - Apr 2019)

B.Sc. in Physics
SUST, Bangladesh

SKILLS

Core Paradigms

- Computer Vision, Image & Video Segmentation, Multi-Object Tracking, Machine Learning

Models & Architectures

- SAM / SAM2 / SAM3, MatSAM, U-Net, Visual Prompt Engineering

Frameworks & Libraries

- PyTorch, OpenCV, SciPy, Scikit-Image, NumPy, Matplotlib

HPC & Compute

- Linux/Ubuntu, Multi-GPU Workflows, CUDA, Git/GitHub, LaTeX

Other Methodologies

- Molecular Dynamics (LAMMPS, ReaxFF), EBSD Analysis, Napari

SELECTED PRESENTATIONS

Poster Presentation: Atomic-scale investigation of the formation of Al nanoclusters in the Pulsed Laser Ablation Technique.

SELECTED COURSEWORK & TRAINING

Deep Learning and Physics – *University of Cologne* (Feb 2026)

Deep Learning with PyTorch: Image Segmentation – *Coursera Project* (Oct 2025)

Computational Many-Body Physics – *University of Cologne* (Feb 2025)

Scientific Programming with Python – *University of Cologne* (Jul 2023)

Google Data Analytics Professional Certificate – *Coursera* (Sep 2022)

LANGUAGES

- English — Fluent (C1)
- German — A1.2 (currently attending a German course)
- Bangla — Native